The Central Limit Theorem: Gamma Distribution

Brandon Johnson and Matthew Conger

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This study investigates the applicability of the Central Limit Theorem (CLT) when sampling from a gamma distribution, characterized by its positive, often skewed nature, using sample sizes ranging from 3 to 101. Through simulations, we generated sample means from gamma distributions with varying shape (k) and scale ( ) parameters, assessing their normality with Shapiro-Wilk, Anderson-Darling, and Lilliefors tests. Our analysis, employing ANOVA and estimated marginal means, reveals that for k 1, the CLT holds reliably when sample size n 30, with sample means approximating a normal distribution. However, for k < 1, where the gamma distribution is highly skewed, significantly larger sample sizes are required, indicating limitations in the CLT's "rule of thumb" for such cases.

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Introduction

In this project, we investigate how well the Central Limit Theorem holds when sampling from a gamma distribution, which is continuous, strictly positive, and often skewed. The gamma distribution is commonly used to model real-world processes like waiting times or rainfall, where values cannot be negative and tend to be unevenly distributed. Because the gamma distribution can look very different from a normal distribution, it provides a useful way to test how sample size and distribution qualities affect the Central Limit Theorem in practice.

To do this, we generate repeated random samples of varying sizes from gamma distributions with different shape and scale parameters. For each set of samples, we calculate the sample means and then use multiple normality tests to assess how close the distribution of these sample means is to a normal distribution. These tests provide different perspectives on normality, with some focusing more on the center of the distribution and others emphasizing the tails.

By comparing the results across different combinations of sample size, shape, and scale, we can get a clearer picture of when the Central Limit Theorem applies - that is, when the sample sample means start behaving like they come from a normal distribution. This helps us make informed recommendations about how large a sample needs to be, depending on how skewed or variable the original gamma distribution is, in order for the Central Limit Theorem to apply reliably.

For this report, the null and alternative hypotheses are consistent with the common "rule of thumb" that the Central Limit Theorem holds when sample size (n) is 30 or greater.:

- **Null Hypothesis ($\mathbf{H_0}$):** Sample means from data drawn from a gamma distribution with given parameters are normally distributed when $n = 30$.
- **Alternative Hypothesis ($\mathbf{H_A}$):** Sample means from data drawn from a gamma distribution with given parameters are not normally distributed when $n = 30$.

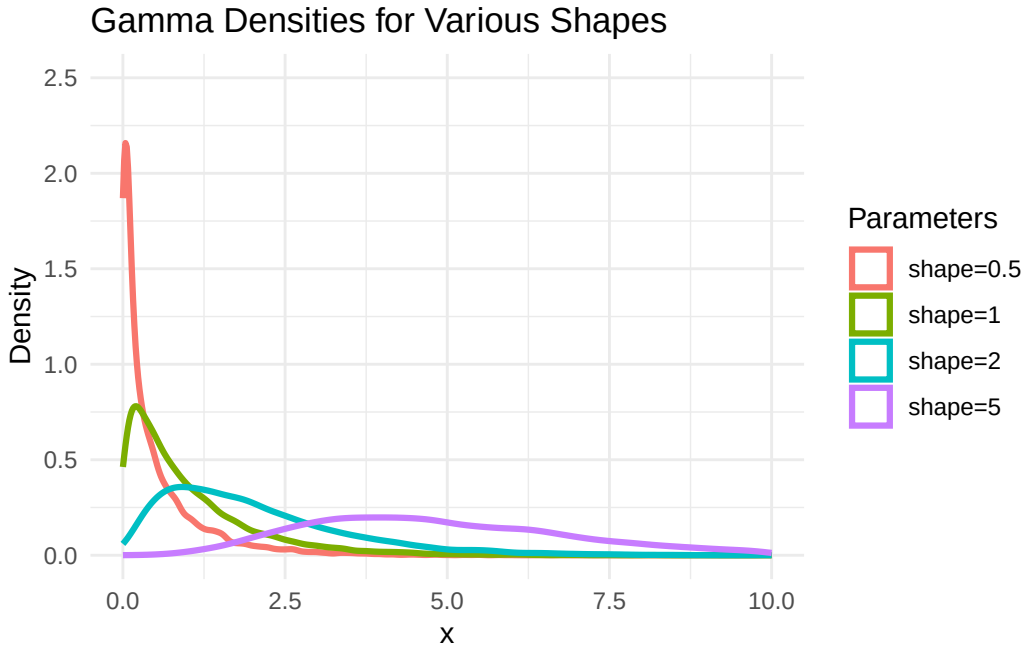
Background

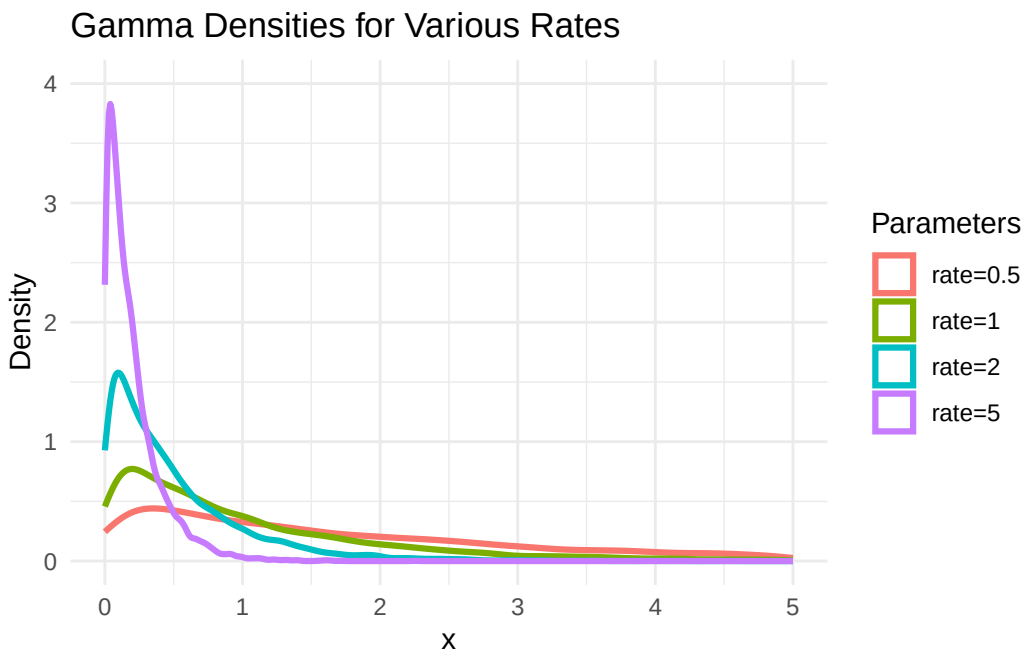
Central Limit Theorem

The Central Limit Theorem (CLT) is a fundamental principle in statistics, stating that the sampling distribution of the mean will approximate a normal distribution as the sample size increases, regardless of the shape of the original population distribution. The general "rule of thumb" is that the CLT holds when sample size (n) is 30 or greater. This theorem is essential in statistical inference, as it forms the foundation for techniques such as hypothesis testing. Understanding how the CLT behaves under different conditions is key to determining how much data is necessary for reliable statistical conclusions.

Gamma Distribution

The gamma distribution is a continuous probability distribution used to model positive, skewed data, such as waiting times, failure rates, or certain physical quantities like rainfall. It is characterized by two parameters: a shape parameter (k or α) and a scale (θ) or rate parameter (β=$1/\theta$). A gamma distribution is typically skewed, especially when the shape parameter is small. As k increases, the distribution goes from resembling an exponential distribution to a bell-shaped distribution. The mean is $k\theta$ (or α/β) and the variance is $k\theta^2$ (or α/β^2). The following graphs show the effects of varying levels of shape and scale on the gamma distribution.





Shapiro-Wilkes Test

The Shapiro-Wilkes Test is a statistical test used to determine whether a data set is normally distributed. The test outputs a test statistic W, which ranges between 0 and 1. A W value closer to 1 suggests that the data is more consistent with a normal distribution.

Anderson Darling Test

The Anderson-Darling test is another test for normality, but it places more emphasis on the tails of the distribution than the Shapiro-Wilkes test. This makes it particularly sensitive to deviations from normality that occur in the extreme values. It also returns a test statistic and a p-value.

Lilliefors Test

The Lillifors test is a variation of the Kolmogorov-Smirnov test, designed to test for normality when the population mean and variance are not known and must be estimated from the sample. Unlike the original K-S test, which assumes a fully specified distribution, the Lillifors test adjusts for this estimation, making it more appropriate for real-world data. It is especially useful in cases where parametric assumptions may not hold.

All three of these tests for normality follow the same general null and alternative hypotheses:

- **Null Hypothesis ($\mathbf{H_0}$):** The data follows a normal distribution.
- **Alternative Hypothesis ($\mathbf{H_A}$):** The data does not follow a normal distribution.

Methodology

Data Generation

Data for this report is generated using a simulation-based approach that models the behavior of sample means drawn from a gamma distribution. To generate this data, we employ a two-step process. First, a specified number of sample means is created by repeatedly drawing n random numbers from gamma distributions with unique k and $/theta$ values using the `rgamma()` function in R. This process is repeated N times. For each repetition, the mean of these values is computed and stored.

After collecting these sample means, the Shapiro-Wilkes, Anderson Darling, and Lilliefors tests are applied to assess whether the resulting sample distributions align with a normal distribution (`shapiro.test()`, `ad.test()`, `lillie.test()`). The test statistics and p-values are recorded for each of these tests.

This entire process is expanded by iterating all combinations of n, k, and θ, by the number of experiments (N_{Exp}). For this report:

- **Sample Sizes (n):** A vector of all prime numbers ranging from 3 to 101.
- **Number of Samples (N):** 30
- **Gamma Distribution Shape (k):** A vector of numbers containing 0.5 and 1 to 10 by 1.
- **Gamma Distribution Scale (θ):** A vector of numbers containing 0.5 and 1 to 10 by 1.
- **Number of Experiments (N_{Exp}):** 1,000

After the data is collected, we interpret the p-values of each normality test using the common level of significance of 5% ($\alpha = 0.05$). We then group the data by each level of n, k, and θ and calculate the proportion of sample distributions that were normal. This proportion is later used in our analysis.

Analysis

To analyze this data, we first perform an exploratory data analysis (EDA) on selected k and θ values to observe overall trends in the generated data.

We then conduct an analysis of the variance test (ANOVA) using the proportion of normal sample distributions as the response and sample size (n), shape (k), and scale (θ) as the factors. We also examine the interactions between each of these factors. The goal of this analysis is to examine the predictors of normality of the sample distributions.

The ANOVA model is used to fit the data:

$$Proportion_{ijk} = \mu + \alpha_i + \beta_j + \lambda_k + (\alpha\beta)_{ij} + (\alpha\lambda)_{ik} + (\beta\lambda)_{jk} + \epsilon_{ijk}$$

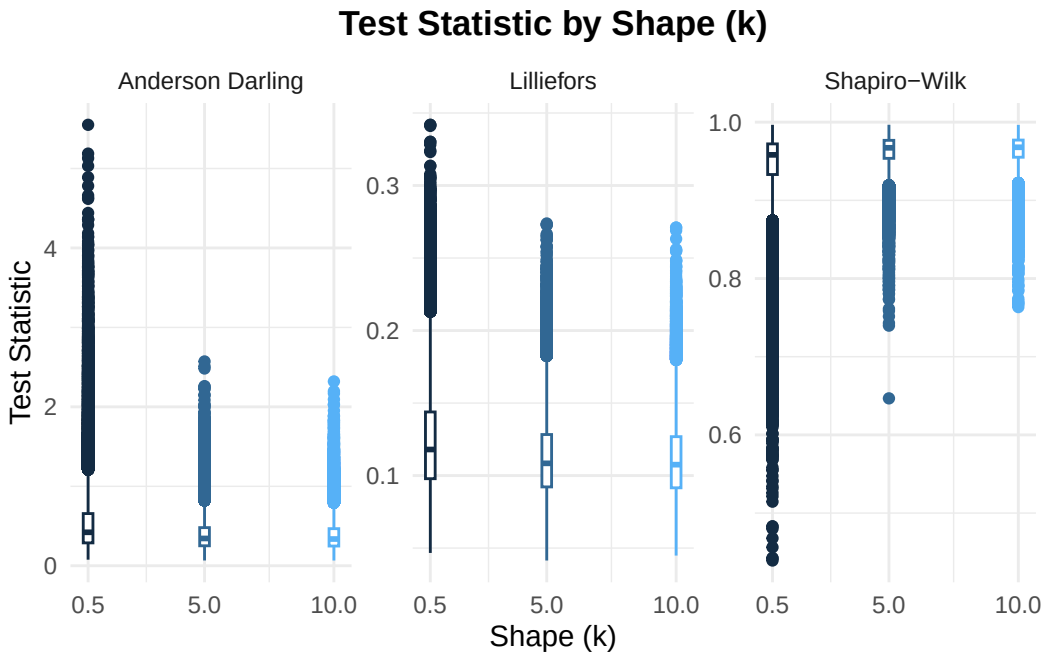
Where:

- $Proportion_{ijk}$ is the observed proportion for the ith level of n, the jth level of k, and the kth level of θ.
- μ is the overall mean.
- α_i is the main effect of n.
- β_j is the main effect of k.
- λ_k is the main effect of θ.
- $(\alpha\beta)_{ij}$ is the interaction effect of n and k.
- $(\alpha\lambda)_{ik}$ is the interaction effect of n and θ.
- $(\beta\lambda)_{jk}$ is the interaction effect of k and θ.
- ϵ_{ijk} is the residual error.

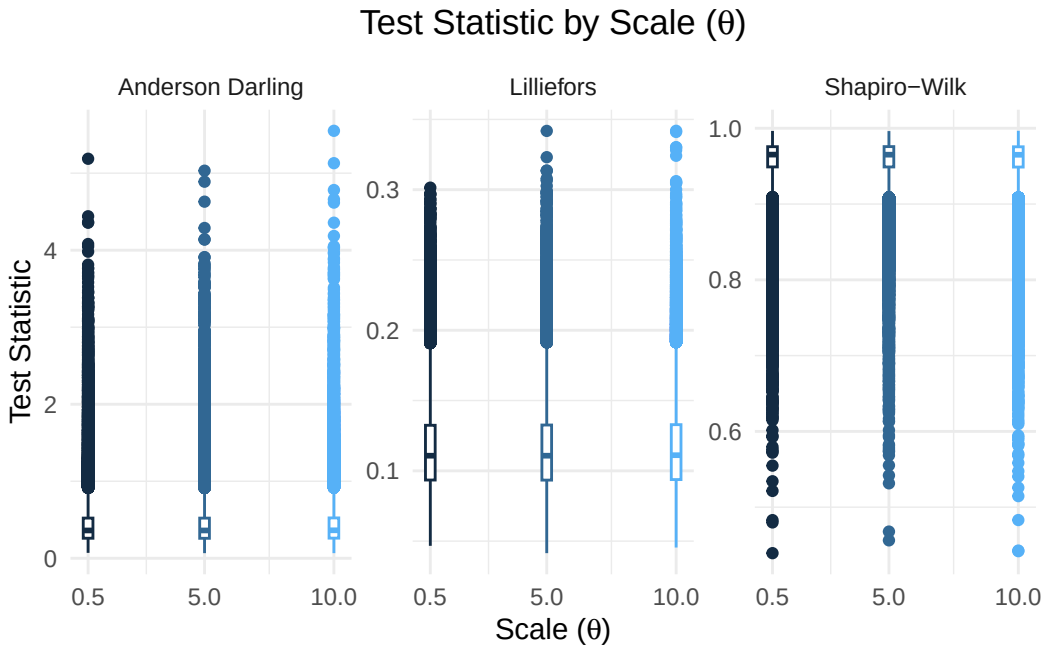
After performing this analysis, we examine pairwise comparisons using Tukey's method as well as a comparison of estimated marginal means (`emmeans()`) to determine which levels of the variables are significant and their effect on the proportion of normal sample distributions. This process is designed to make overall general recommendations for sample size for the CLT to hold when analyzing data from a gamma distribution.

Exploratory Data Analysis

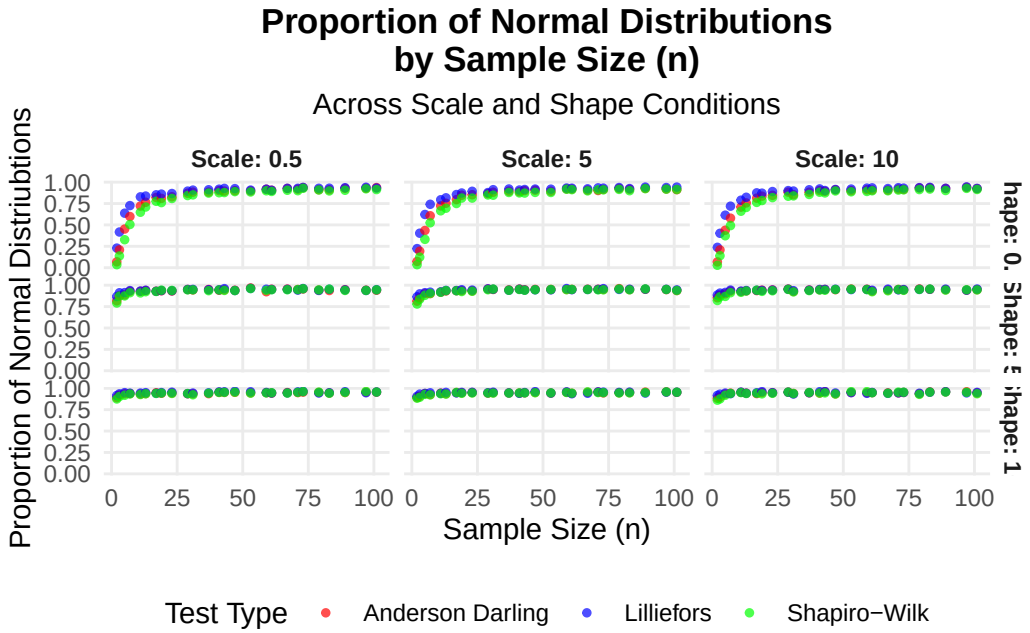
Boxplots of Test Statistics by Gamma Distribution Parameters



These boxplots display the distribution of test statistics for the three tests for normality across three levels of the k and θ parameters. The plots reveal that as the shape and scale parameters increase, the spread and central tendency of the test statistics shift. The Anderson Darling test statistics show higher variability, the Lilliefors statistics maintain a tighter range, and the Shapiro-Wilkes W statistics cluster closer to 1. This highlights the slight differences among each test for normality that is represented. It is important to note that the y-axis changes for each facet of the graphs.



Scatter Plot of Proportion of Normal Distributions by Sample Size



This scatter plot illustrates the results of the three statistical tests across the varying levels of n.

These results are presented under each combination of the three selected levels of the gamma distribution parameters k and θ. Each subplot shows the proportion of sample distributions identified as normal graphed against sample size n.

The overall results show that as n increases, the proportion of normal sample distributions for all three normality tests approaches 1. The graphs reveal that as shape (θ) increases, the proportion approaches 1 regardless of n. This trend is not observed in scale (k) which seems to have no effect on the proportion of normal sample distributions despite its variation. This graph suggests that sample size and shape may have significant effects on the proportion, while scale likely does not. These relationships are further explored in the following section.

Results

ANOVA Analysis

The results of the ANOVA mirror the suggestions from the second exploratory data analysis. Specifically, the analysis revealed significant effects of n and k on the proportion of normal sample distributions. Furthermore, the interaction between these two variables was also significant. As expected, θ did not have a significant effect on the proportion nor did any of the interactions involving θ. The following table displays the ANOVA results.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(shape)	10	16.1357690	1.6135769	2532.8518346	0.0000000
factor(rate)	10	0.0011743	0.0001174	0.1843288	0.9974008
factor(n)	25	27.5024508	1.1000980	1726.8376306	0.0000000
factor(shape):factor(rate)	100	0.0161961	0.0001620	0.2542316	1.0000000
factor(shape):factor(n)	250	33.3127316	0.1332509	209.1656444	0.0000000
factor(rate):factor(n)	250	0.0367518	0.0001470	0.2307592	1.0000000
factor(shape):factor(rate):factor(n)	2500	0.3665306	0.0001466	0.2301391	1.0000000
Residuals	6292	4.0083773	0.0006371	NA	NA

In response to the insignificant effects of θ, a second model, excluding θ, was analyzed. The following table displays the results of the second ANOVA model. For simplicity, this model is used in the pairwise comparison analysis as the F-statistics are somewhat consistent with the original model.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(shape)	10	16.13577	1.6135769	3334.2415	0
factor(n)	25	27.50245	1.1000980	2273.2059	0

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(shape):factor(n)	250	33.31273	0.1332509	275.3453	0
Residuals	9152	4.42903	0.0004839	NA	NA

Estimated Marginal Means

After performing ANOVA analysis on the data, we analyzed the estimated marginal means of the data, paying specific attention to the values of n at which the proportion of normal sample distributions stabilizes between .9 and .95. This gives us a good idea of the required sample size needed for the CLT to hold when the scale of the gamma distribution from which the data is drawn changes. The following table shows the minimum value of n for each level of k that this threshold is met.

shape	n	emmean
0.5	47	0.9039394
1.0	23	0.9020000
2.0	13	0.9101818
3.0	11	0.9133333
4.0	7	0.9076364
5.0	5	0.9013939
6.0	5	0.9112121
7.0	5	0.9206061
8.0	3	0.9036667
9.0	3	0.9075152
10.0	3	0.9155152

These results indicate that for levels of k between 1 and 10, the general requirement of the CLT that $n \geq 30$ is more than enough to ensure the CLT holds. However, as k falls below 1, and the original gamma distribution resembles an exponential distribution, far larger n values are required for the CLT to hold.

Conclusion

The results of this report provide strong evidence to support the null hypothesis that sample means from a gamma distribution with given parameters are normally distributed when the sample size (n) is 30 or greater, particularly for shape parameters (k) between 1 and 10 For these values of k, the proportion of normal sample distributions, as assessed by three different normality tests, consistently stabilized above 0.9 when $n \geq 30$, indicating that the Central Limit Theorem holds reliably under these conditions.

However, when $k \leq 1$, the gamma distribution resembles a highly skewed exponential distribution and the null hypothesis is rejected in favor of the alternative. Significantly larger sample sizes were required to achieve normality, suggesting that the Central Limit Theorem's "rule of thumb" of $n \geq 30$ is insufficient in such cases.

The results also revealed no effect of the scale parameter (θ) on the required sample size (n) for the Central Limit Theorem to hold.

Implications

The implications of these findings are significant for statistical practice, particularly when dealing with skewed data like waiting times or rainfall, which are often modeled by gamma distributions Failing to reject our null hypothesis for $k \geq 1$ means researches can confidently apply normal-based inferential methods with sample sizes greater than 30. However, for highly skewed distributions ($k \leq 1$), larger sample sizes are necessary to ensure the Central Limit Theorem applies.

Recommendations for Future Research

Future researchers might consider testing additional complex distributions to explore the CLT's limits with varied skewness. Including additional normality tests may offer more robust insights. Finally, the use of real-world datasets, such as rainfall or waiting times, alongside these simulations would strengthen the practical relevance of these findings.

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Appendix

Appendix 1

Tables of Raw Data. First and last five observations are shown.

n	N	shape	rate	p_vl_shap	tstat_shap	p_vl_lil	tstat_lil	p_vl_ad	tstat_ad
2	30	0.5	0.5	0.0001217	0.8144732	0.0003312	0.2289732	0.0002256	1.6611947
2	30	0.5	0.5	0.0001743	0.8226214	0.0001228	0.2399177	0.0000786	1.8430358
2	30	0.5	0.5	0.0000542	0.7954539	0.0012838	0.2129276	0.0000043	2.3463790
2	30	0.5	0.5	0.0000183	0.7684006	0.0003242	0.2292173	0.0000060	2.2864971
2	30	0.5	0.5	0.0055629	0.8926003	0.0918416	0.1480579	0.0202936	0.8875935
2	30	0.5	0.5	0.0005205	0.8463498	0.0327643	0.1667276	0.0011691	1.3778632

	n	N	shape	rate	p_vl_shap	tstat_shap	p_vl_lil	tstat_lil	p_vl_ad	tstat_ad
3145995	101	30	10	10	0.7115831	0.9759798	0.3080237	0.1217555	0.5213869	0.3171736
3145996	101	30	10	10	0.9339926	0.9848147	0.6085444	0.1008473	0.8671107	0.2019626
3145997	101	30	10	10	0.0887437	0.9396024	0.3854732	0.1157351	0.2505462	0.4549480
3145998	101	30	10	10	0.0257085	0.9192902	0.0447229	0.1613882	0.0250121	0.8517461
3145999	101	30	10	10	0.1047171	0.9422701	0.7371229	0.0925244	0.1957983	0.4975724
3146000	101	30	10	10	0.7114965	0.9759769	0.8829936	0.0810876	0.8741714	0.1990271

Appendix 2

Functions for Generating Data.

```
library(nortest)nnnmain <- function(n, N, N_Exp, shape, rate) {n  results<-data.frame(n=integ
```

```
proj1<-function(n, N, N_Exp, shape, rate){
  if(!exists("main", mode = "function")){
    source("Gamma_main.R")
  }
  results<-data.frame(n=integer(),
                      N=integer(),
                      shape=numeric(),
                      rate=numeric(),
                      p_vl_shap=numeric(),
                      tstat_shap=numeric(),
                      p_val_lil=numeric(),
```

```
                    tstat_lil=numeric(),
                    p_val_ad=numeric(),
                    tstat_ad=numeric()
  )
  for (i in N){
    for (j in n){
      for (k in shape){
        for (l in rate){
          results<-rbind(results,main(j,i,N_Exp,k,l))
        }
      }
    }
  }
  results<-data.frame(results)
  return(results)
}
```

Appendix 3

R Code.

```
knitr::opts_chunk$set(echo = TRUE)

library(tidyverse)
library(knitr)
library(emmeans)
knitr::write_bib(c('knitr', 'stringr', 'dplyr', 'tidyverse','nortest','emmeans'), file = 'pa
#Shapes
shapes <- c(0.5, 1, 2, 5)
rate <- 1
param_grid <- expand.grid(shape = shapes)
sampled_data <- param_grid |>
  rowwise() |>
  mutate(data = list(rgamma(10000, shape = shape, rate = rate))) |>
  unnest(cols = c(data)) |>
  rename(x = data) |>
  mutate(group = paste0("shape=", shape))

shapegraph<-ggplot(sampled_data, aes(x = x, color = group)) +
  geom_density(size = 1.1) +
  labs(title = "Gamma Densities for Various Shapes",
       x = "x", y = "Density", color = "Parameters") +
```

```
  xlim(0, 10) +
  ylim(0, 2.5) +
  theme_minimal()


#Rates
shape <- 1
rates <- c(0.5, 1, 2, 5)
param_grid <- expand.grid(rate = rates)
sampled_data <- param_grid |>
  rowwise() |>
  mutate(data = list(rgamma(10000, shape = shape, rate = rate))) |>
  unnest(cols = c(data)) |>
  rename(x = data) |>
  mutate(group = paste0("rate=", rate))

rategraph<-ggplot(sampled_data, aes(x = x, color = group)) +
  geom_density(size = 1.1) +
  labs(title = "Gamma Densities for Various Rates",
       x = "x", y = "Density", color = "Parameters") +
  xlim(0, 5) +
  ylim(0, 4) +
  theme_minimal()
shapegraph
rategraph
if(!exists("proj1", mode = "function")){
  source("Gamma_proj1.R")
}
n<-c(1,2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97,101)
N<-c(30)
N_Exp<-c(1000)
shape<-c(0.5,1,2,3,4,5,6,7,8,9,10)
rate<-c(0.5,1,2,3,4,5,6,7,8,9,10)

start_time <- proc.time()
#testmc2 <- proj1(n,N,N_Exp,shape=shape,rate=rate)
end_time <- proc.time()

timetorun<-start_time - end_time

#We made the decision that we would load in a large data set that has
#already been created by us using this exact code, to reduce render time.
```

```r
#loads data
load("data2.Rda")


#Pivots our data
df_long <- testmc |>
  pivot_longer(
    cols = c(p_vl_shap, tstat_shap, p_vl_lil, tstat_lil, p_vl_ad, tstat_ad),
    names_to = c(".value", "test"),
    names_pattern = "(p_vl|tstat)_(shap|lil|ad)"
  )


#filters down our data to be more managable
df_long2 <- df_long|>
  filter(shape %in% c(0.5,5,10))|>
  filter(rate %in% c(0.5,5,10))|>
  mutate(normal=(p_vl>=0.05))


#new table with each combination of factors with a proportion of trues
df_long3<-df_long2|>
  group_by(n,test,shape,rate)|>
  summarize(prop=mean(normal))

#Test stat vs shape
test_labels <- c("ad" = "Anderson Darling",
                 "lil" = "Lilliefors",
                 "shap" = "Shapiro-Wilk")
ggplot(df_long2, aes(x = shape, y = tstat, group = shape, color = shape)) +
  geom_boxplot(show.legend = F) +
  facet_wrap(~test, scales = "free_y", labeller = labeller(test = test_labels)) +
  labs(
    title = "Test Statistic by Shape (k)",
    x = "Shape (k)",
    y = "Test Statistic"
  ) +
  scale_x_continuous(breaks = c(0.5, 5, 10)) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = .5, face = "bold"))
#Test stat vs scale
ggplot(df_long2, aes(x = rate, y = tstat, group = rate, color = rate)) +
```

```
  geom_boxplot(show.legend = F) +
  facet_wrap(~test, scales = "free_y", labeller = labeller(test = test_labels)) +
  labs(
    title = expression(paste("Test Statistic by Scale (", theta, ")")),
    x = expression(paste("Scale (", theta, ")")),
    y = "Test Statistic"
  ) +
  scale_x_continuous(breaks = c(0.5, 5, 10)) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = .5, face = "bold"))
#Grid of Tables
df_long3_plot <- df_long3 %>%
  mutate(Scale = rate,
         Shape = shape)

ggplot(df_long3_plot, aes(x = n, y = prop, color = test)) +
  geom_point(size = 1, alpha = 0.7) +
  facet_grid(Shape ~ Scale,
             labeller = label_both) +
  labs(
    title = "Proportion of Normal Distributions\nby Sample Size (n)",
    subtitle = "Across Scale and Shape Conditions",
    x = "Sample Size (n)",
    y = "Proportion of Normal Distriubtions",
    color = "Test Type"
  ) +
  scale_color_manual(
    name = "Test Type",
    values = c("ad" = "red", "lil" = "blue", "shap" = "green"),
    labels = c("ad" = "Anderson Darling",
                                "lil" = "Lilliefors",
                                "shap" = "Shapiro-Wilk")) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    plot.subtitle = element_text(hjust = .5),
    strip.text = element_text(face = "bold"),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

df_long4 <- df_long|>
```

```
  mutate(normal=(p_vl>=0.05)) %>%
  group_by(n,test,shape,rate)|>
  summarize(prop=mean(normal))

mcmod<-aov(prop~factor(shape)*factor(rate)*factor(n), data=df_long4)

kable(anova(mcmod))
mcmod2<-aov(prop~factor(shape)*factor(n), data=df_long4)
kable(anova(mcmod2))
mean2<-as.data.frame(emmeans(mcmod2, ~ shape*n,adjust="tukey"))
mean2_filtered <- mean2 %>%
  filter(emmean >= 0.9) %>%
  group_by(shape) %>%
  slice(which.min(n)) %>%
  ungroup() %>%
  select(shape, n, emmean)

  kable(mean2_filtered)
kable(head(testmc))
kable(tail(testmc))
cat(readLines("Gamma_main.R"), sep = "n")
cat(readLines("Gamma_proj1.R"), sep = "\n")
```